

SIMATS ENGINEERING
CO3 - ASSESSMENT TOOL - 1: Experiments

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

Name	P. Guru
Register Number	192521010

COURSE OUTCOME COVEREDCO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Assessment Tool Weightage: 40%

Dave's Taxonomy: Precision (Level 3)
Question Count: 4

Total Marks: 40

Duration : 60 Minutes

Code of Conduct

I P. Guru /192521010 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P. Guru

Experiments Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Experimental Design	25%	Design is systematic and technically strong	Mostly correct design	Basic design with gaps	Poorly designed activity
Use of Tools / Platforms	20%	Appropriate and effective use of cloud tools	Good tool usage with minor issues	Limited tool usage	Incorrect or missing tool usage
Application of Concepts	20%	Concepts are applied accurately to practical tasks	Mostly accurate application	Partial application	Weak application
Analysis and Output	20%	Strong analysis and meaningful outcome interpretation	Good analysis with minor gaps	Basic analysis	Little or no analysis
Documentation	15%	Clear, structured, and complete documentation	Mostly complete documentation	Partially complete	Poor documentation

Total Marks Awarded:

Assessment Tasks

Experiments

Experiment 1: Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

Experiment 2: Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org , over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

Experiment 1: Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

AIM: To Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS

PROCEDURE:

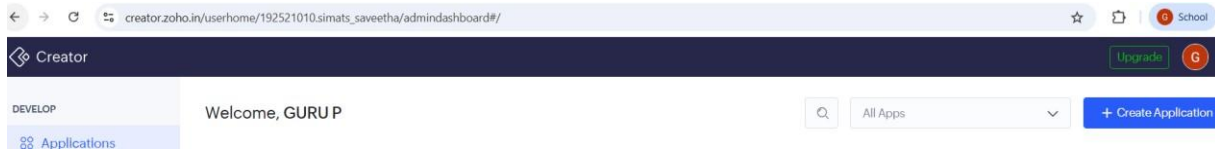
- step1: Go to zoho.com.
- step 2: Log into the zoho.com.
- step 3: Select one application step.
- step4: Enter application name as library book reservation system.
- step 5: Created new application as library book reservation system.
- step 6: Select one form
- step 7: The software has been created.

IMPLEMENTATION:

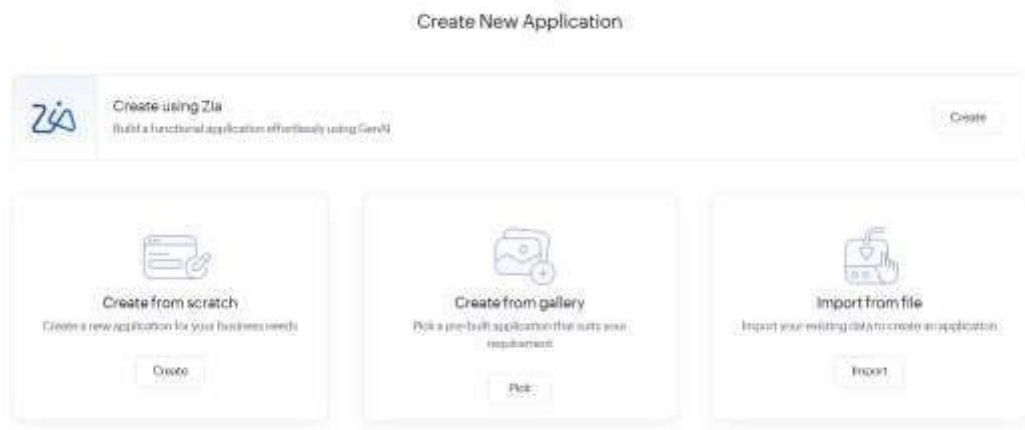
STEP1: GOTO ZOHO.COM



STEP 2: LOGINTOTHE ZOHO.COM



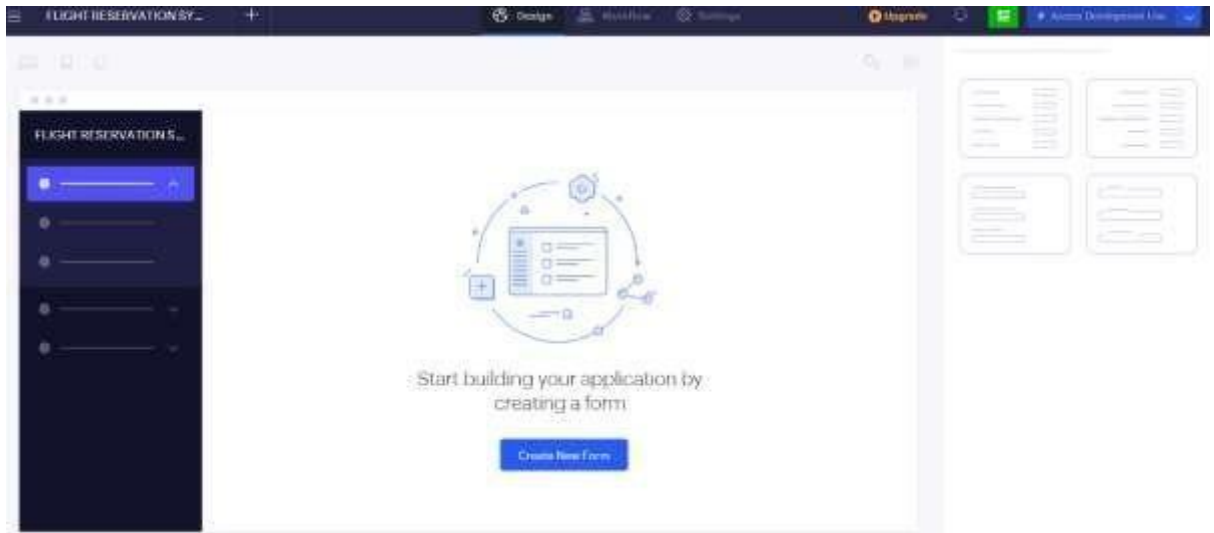
STEP 3: SELECT ONE APPLICATION



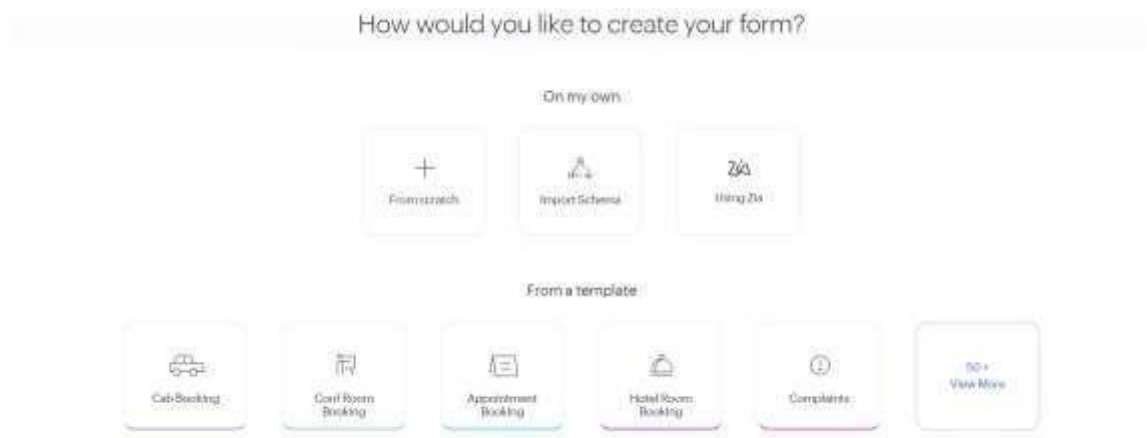
STEP 4: ENTER APPLICATION NAME



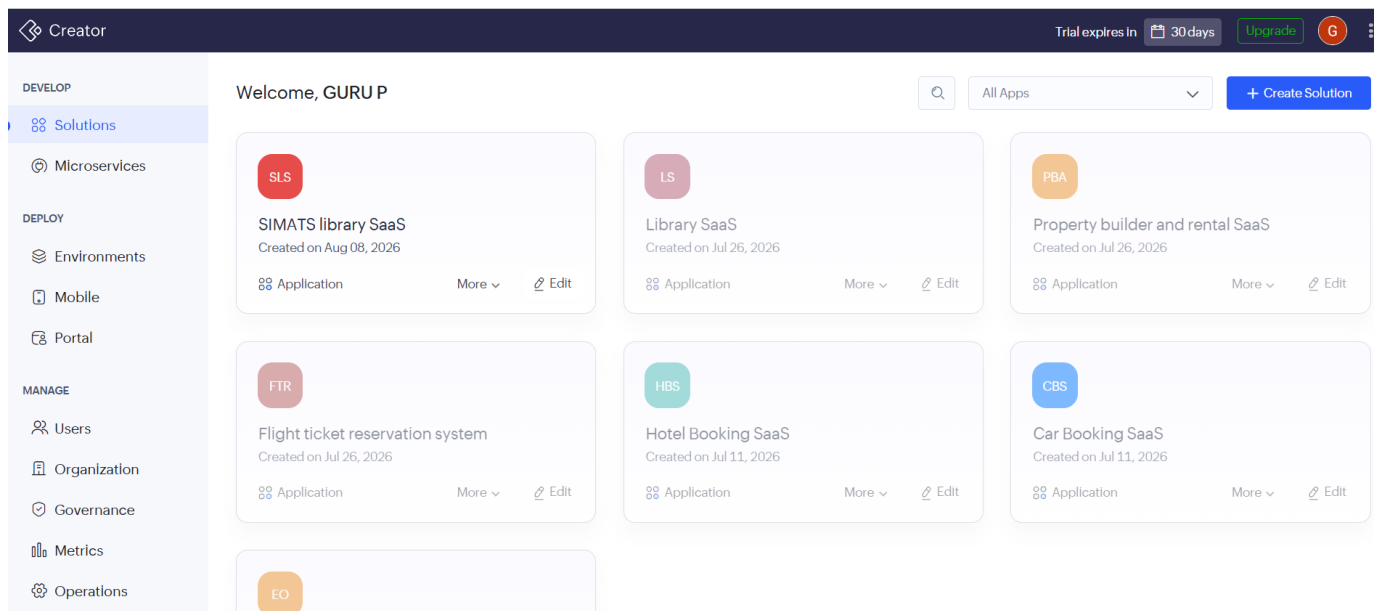
STEP 5: CREATED NEW APPLICATION



STEP 6: SELECT ONE FORM



STEP 7: THE SOFTWARE HAS BEEN CREATED.



The screenshot displays the SIMATS library SaaS application interface. On the left is a dark sidebar with navigation links: Authors, Publishers, Books, Copies, Members (highlighted), All Members, Reservations, and Loans. The main content area is titled 'Members' and contains a form with the following fields: 'Full name' (text input), 'Email' (text input), 'Phone' (text input with a placeholder '#####'), and 'role' (dropdown menu with '-Select-' and a downward arrow). A blue 'Submit' button is located below the form. The top of the browser window shows the title 'SIMATS library SaaS' and some window control icons. The bottom of the sidebar shows a user profile icon and the name 'GURU P'.

Experiment 2: Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org , over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

AIM:

To create a simple cloud software application for **Payroll Processing System** using **Zoho Creator** as a Cloud Service Provider to demonstrate the **Software as a Service (SaaS)** model.

PROCEDURE:

Step 1: Go to **zoho.com**.

Step 2: Log in to your Zoho account.

Step 3: Open **Zoho Creator** from the Zoho applications.

Step 4: Click **Create New Application**.

Step 5: Enter the application name as **Payroll Processing System**.

Step 6: Create a new application.

Step 7: Create a new form named **Employee Payroll Details**.

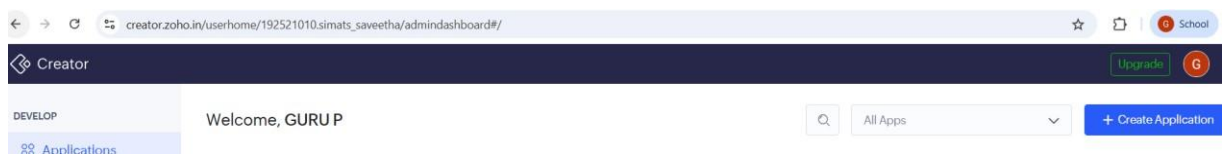
Step 8: The **Payroll Processing System** software has been successfully created and deployed on the Zoho cloud platform.

IMPLEMENTATION:

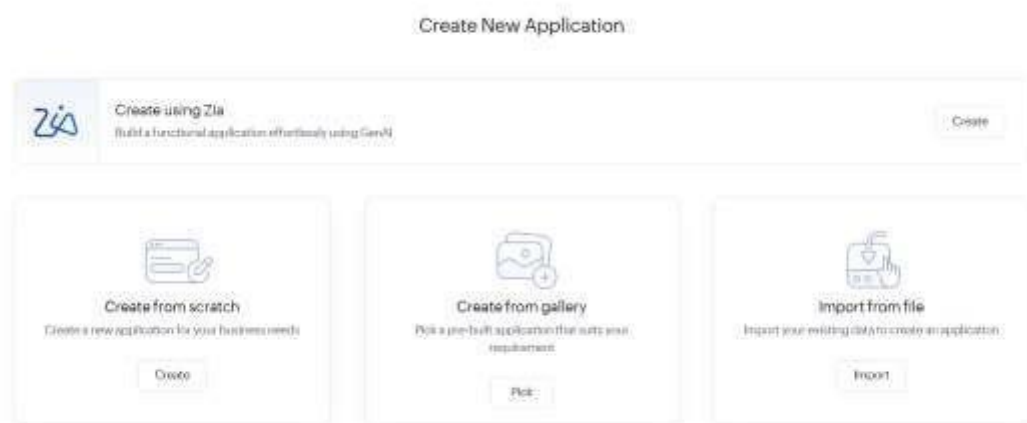
STEP1: GOTO ZOHOCOM



STEP 2: LOGINTOTHE ZOHOCOM



STEP 3: SELECT ONE APPLICATION



STEP 4: ENTER APPLICATION NAME

Create from scratch

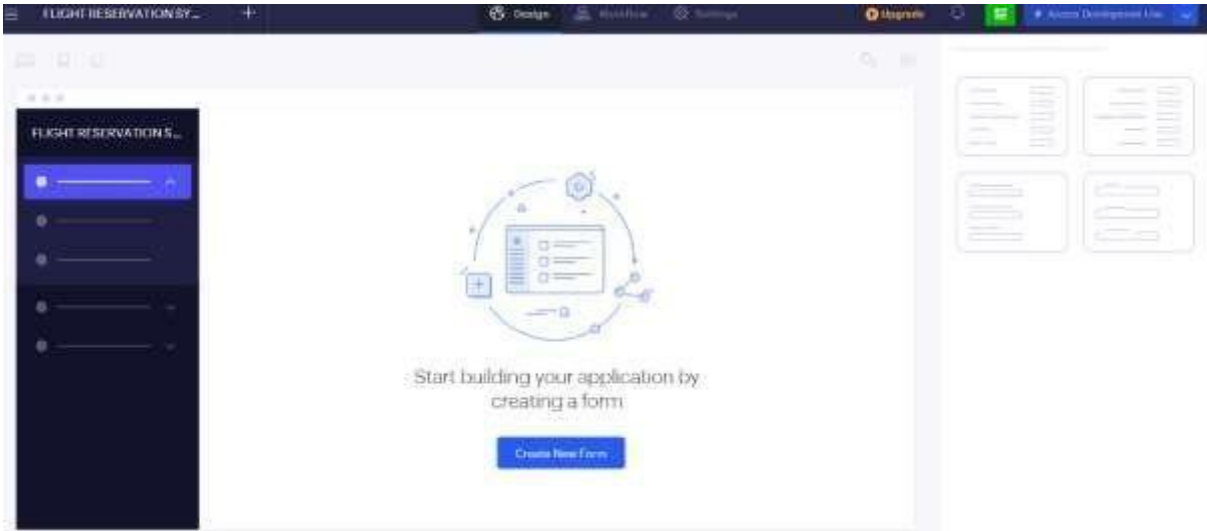
Application Name *

Enter your application name

☐ Enable Environment (Learn More)

Create Application

STEP 5: CREATED NEW APPLICATION



STEP 6: SELECT ONE FORM

How would you like to create your form?

On my own

+

From scratch

📄

Import Schema

🔗

Using Zia

From a template

🚗

Cab Booking

🏠

Guest Room Booking

📅

Appointment Booking

🏨

Hotel Room Booking

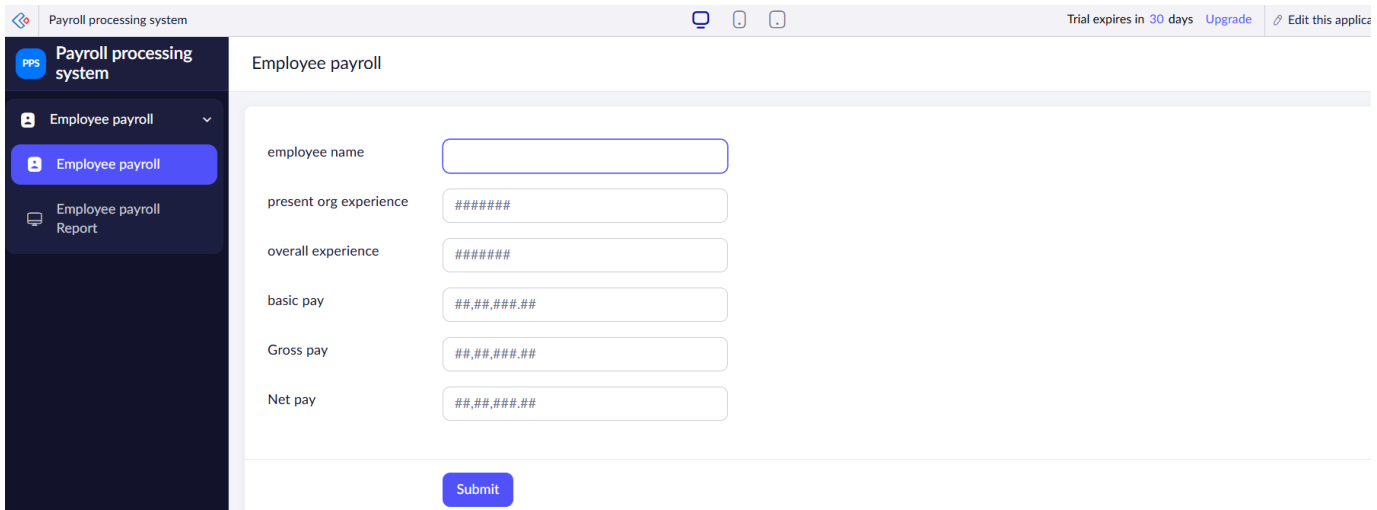
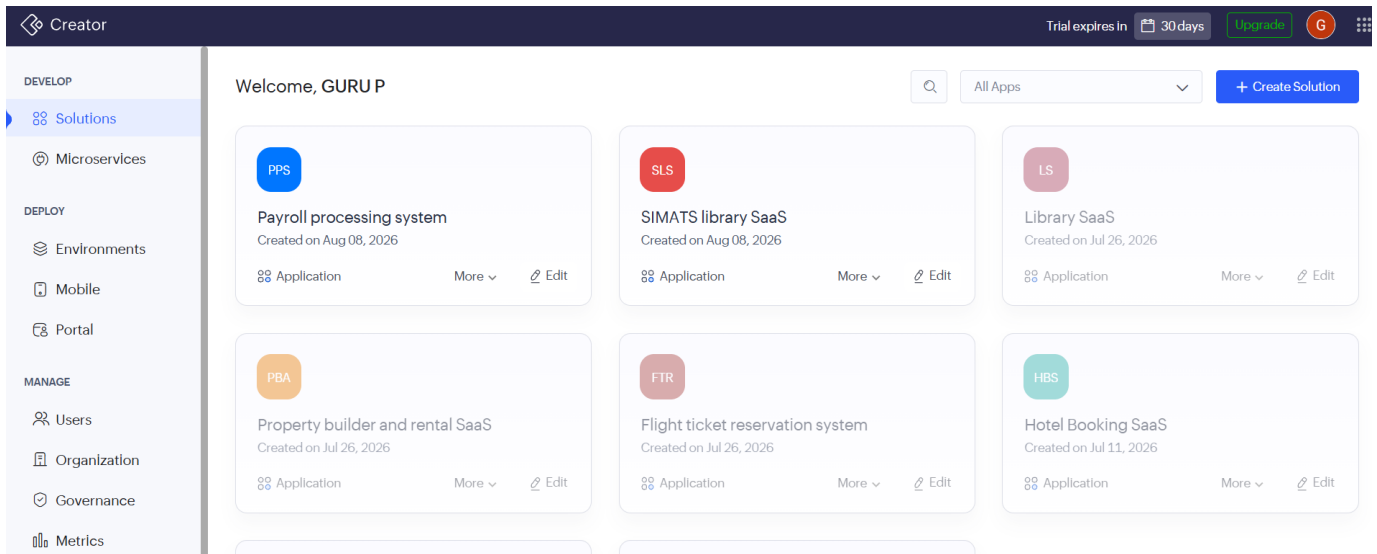
🕒

Complaints

50+

View More

STEP 7: THE SOFTWARE HAS BEEN CREATED.



Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

AIM:

To demonstrate virtualization by installing type-2 hypervisor in your device, create and configure VM image with a host operating system (either windows/linux).

PROCEDURE:

STEP 1:Download VMware workstation and installed as type 2hypervisor.

STEP2:Download ubuntu or tiny OS as iso image file.

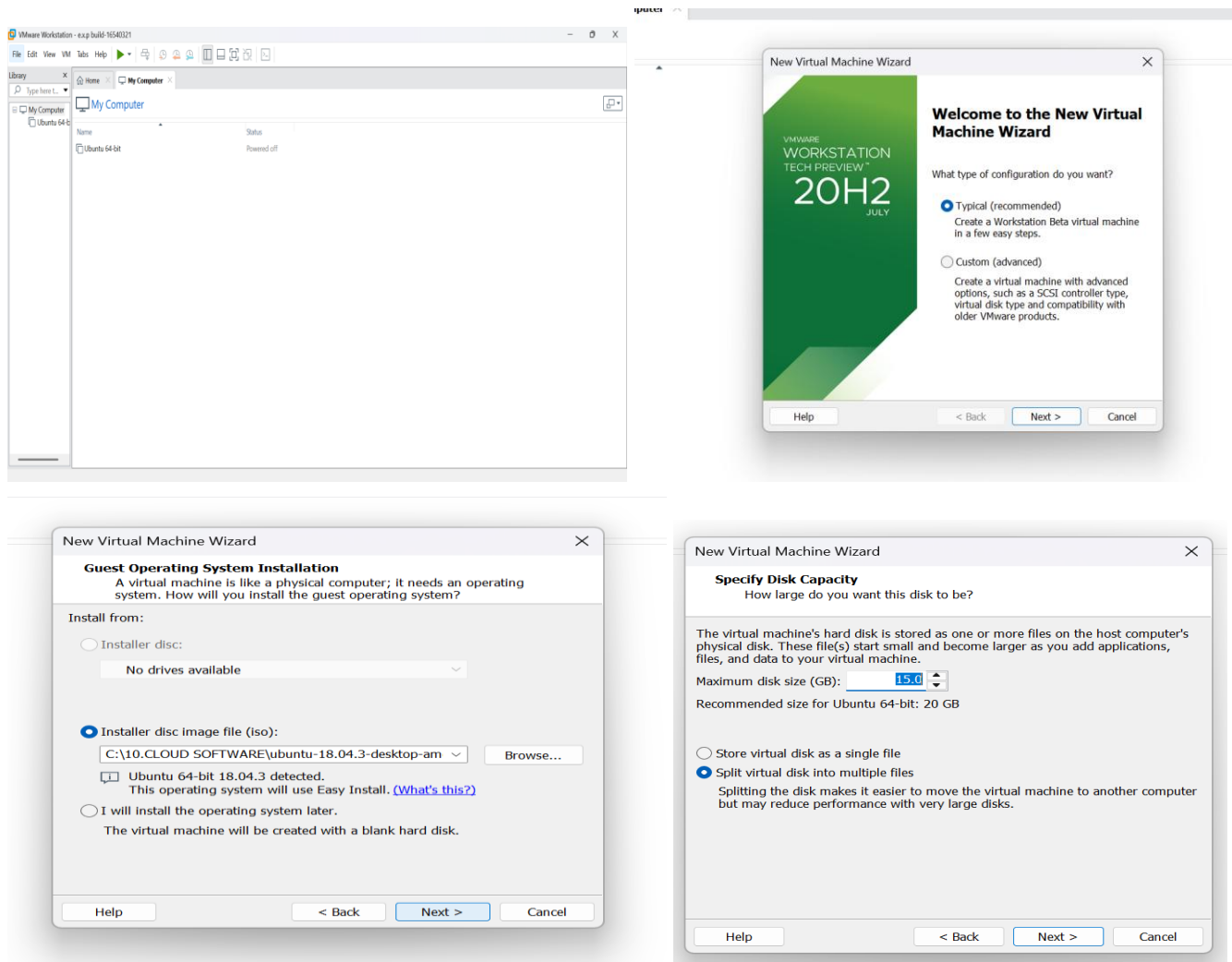
STEP 3: In VMware workstation->create new VM.

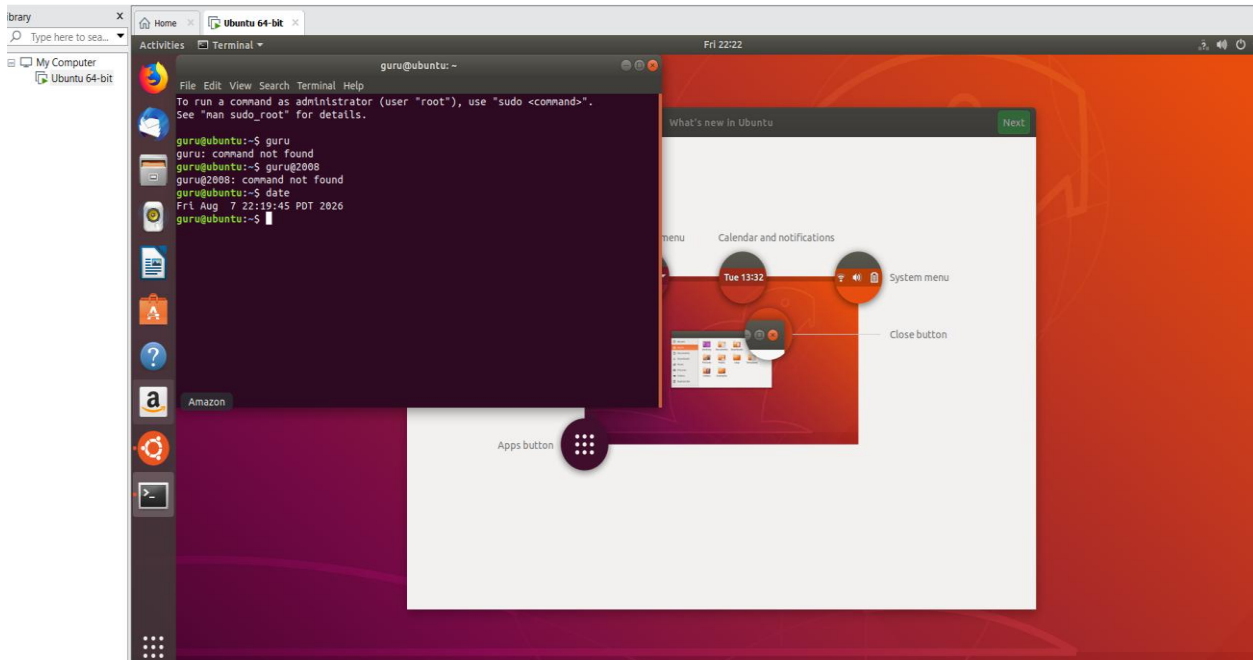
STEP 4: Do the basic configuration settings.

STEP 5: Created tiny OS virtual machine.

STEP 6: Launch the VM.

IMPEMENTATION:





Result:

The Ubuntu virtual machine was successfully created using VMware Workstation.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

Aim

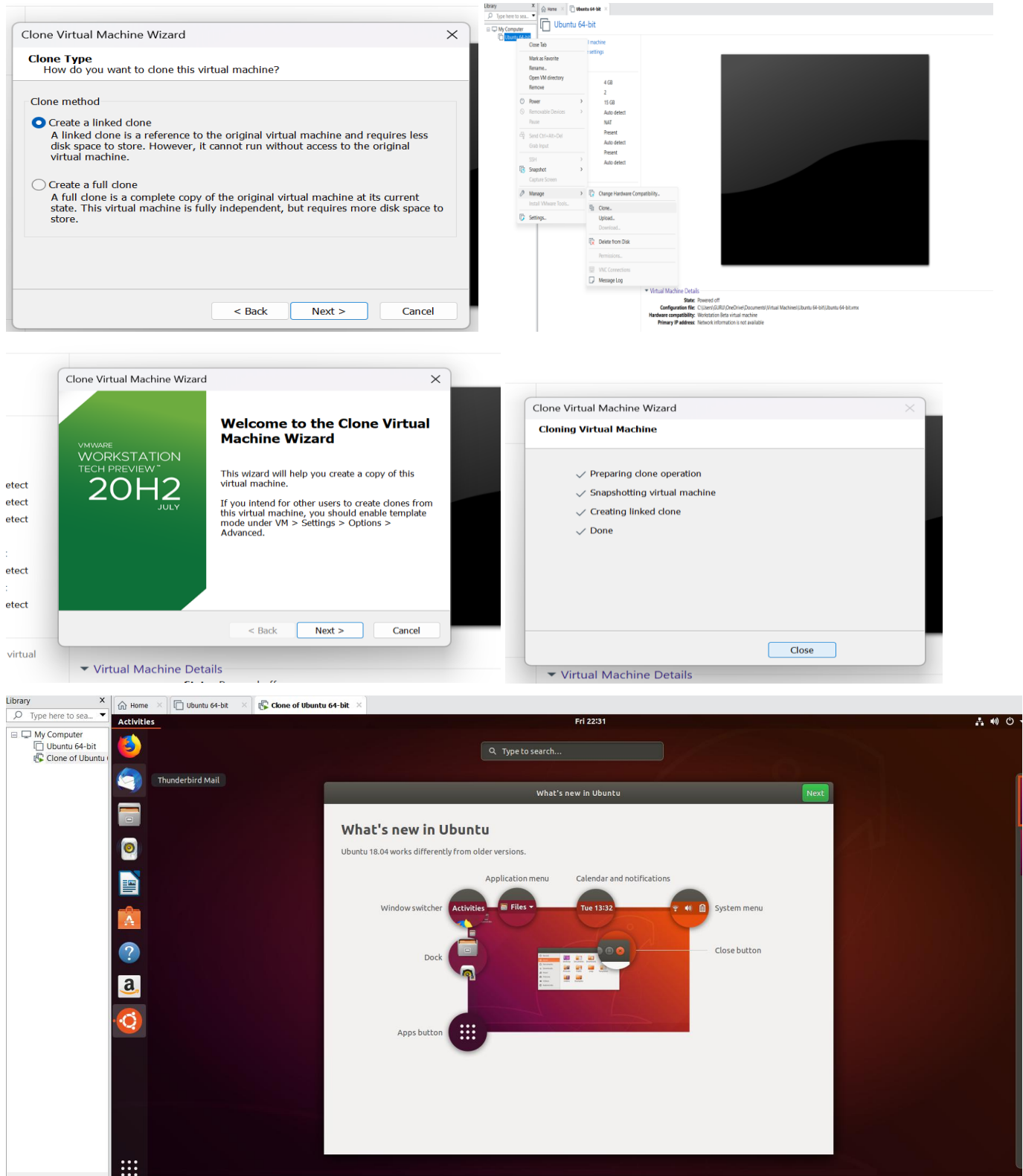
To demonstrate virtualization using VMware Workstation (Type-2 Hypervisor) by creating a virtual machine, cloning it, and testing the cloned VM.

Procedure

1. Install and open VMware Workstation on the Windows host system.
2. Create a new virtual machine and install Ubuntu Linux as the guest OS.
3. Configure the VM with suitable RAM, processor, storage, and network settings.
4. Shut down the original VM and create a copy of its VM folder.
5. Add the copied VM to VMware Workstation and name it Ubuntu Clone.
6. Select "I Copied It" when VMware asks about the VM identity.

7. Start the cloned VM and verify that Ubuntu and the previous files/configuration are loaded successfully.

Implementation:



Result :

The Ubuntu virtual machine was successfully created using VMware Workstation, and a cloned VM was created and tested successfully, demonstrating Type-2 virtualization and VM cloning.